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2.80s. Calculate (i) the mean value of the period of oscillation (ii) the absolute error in each measurement (iii) the mean absolute error (iv) the relative error (v) the percentage error. Express the result in proper form.

Solution

$t_1 = 2.63\,s,\ t_2 = 2.56\,s,\ t_3 = 2.42\,s,$
$t_4 = 2.71\,s,\ t_5 = 2.80\,s$

(i) $T_m = \dfrac{t_1 + t_2 + t_3 + t_4 + t_5}{5}$

$= \dfrac{2.63 + 2.56 + 2.42 + 2.71 + 2.80}{5}$

$T_m = \dfrac{13.12}{5} = 2.624s$

$T_m = 2.62s$ (Rounded off to 2nd decimal place)

(ii) Absolute error $\Delta T = T_m - t$

$\Delta T_1 = 2.62 - 2.63 = -0.01s$
$\Delta T_2 = 2.62 - 2.56 = +0.06s$
$\Delta T_3 = 2.62 - 2.42 = +0.20s$
$\Delta T_4 = 2.62 - 2.71 = -0.09s$
$\Delta T_5 = 2.62 - 2.80 = -0.18s$

(iii) Mean absolute error $= \dfrac{\Sigma|\Delta T_i|}{n}$

$\Delta T_m = \dfrac{0.01 + 0.06 + 0.20 + 0.09 + 0.18}{5}$

$\Delta T_m = \dfrac{0.54}{5} = 0.108s = 0.11s$ (Rounded off to 2nd decimal place)

(iv) Relative error:

$S_T = \dfrac{\Delta T_m}{T_m} = \dfrac{0.11}{2.62} = 0.0\ 419$

$S_T = 0.04$

(v) Percentage error in T = 0.04 × 100% = 4%

(vi) Time period of simple pendulum = T = (2.62 ± 0.11)s

1.6.4 Propagation of errors

A number of measured quantities may be involved in the final calculation of an experiment. Different types of instruments might have been used for taking readings. Then we may have to look at the errors in measuring various quantities, collectively.

The error in the final result depends on

(i) The errors in the individual measurements

(ii) On the nature of mathematical operations performed to get the final result. So we should know the rules to combine the errors.

The various possibilities of the propagation or combination of errors in different mathematical operations are discussed below:

(i) Error in the sum of two quantities

Let ΔA and ΔB be the absolute errors in the two quantities A and B respectively. Then,

Measured value of A = A ± ΔA

Measured value of B = B ± ΔB

Consider the sum, Z = A + B